Lecture 12: Introduction to Instrumental Variables

Jeremy Majerovitz (University of Notre Dame)

Econometrics, Fall 2026

Returns to Capital (de Mel, McKenzie, and Woodruff 2008)

We previously discussed de Mel, McKenzie, and Woodruff (2008). They combined the grants into a single instrument, Grant Amount$_{it}$, which was equal to the total number of rupees that the microenterprise had received in grants up until time t. They ran the IV regression:

$$\text{Profit}_{it} = \alpha + \beta \cdot \text{Capital}_{it} + \gamma_t + \varepsilon_{it}$$

with the first-stage regression:

$$\text{Capital}_{it} = \theta + \pi \cdot \text{Grant Amount}_{it} + \delta_t + u_{it}$$

They find extremely high returns to capital, averaging 5% per month (about 60% per year).

Exercise 1. What does it mean for the instrument to be relevant? How can we test for relevance?

Exercise 2. Is the grant amount as-good-as-randomly assigned? Why or why not?

Returns to Capital (de Mel, McKenzie, and Woodruff 2008)

We previously discussed de Mel, McKenzie, and Woodruff (2008). They combined the grants into a single instrument, Grant Amount$_{it}$, which was equal to the total number of rupees that the microenterprise had received in grants up until time t. They ran the IV regression:

$$\text{Profit}_{it} = \alpha + \beta \cdot \text{Capital}_{it} + \gamma_t + \varepsilon_{it}$$

with the first-stage regression:

$$\text{Capital}_{it} = \theta + \pi \cdot \text{Grant Amount}_{it} + \delta_t + u_{it}$$

They find extremely high returns to capital, averaging 5% per month (about 60% per year).

Exercise 3. What would the "reduced form" regression be?

Exercise 4. What does the exclusion restriction mean in this setting? Do you think the grant amount is excludable? Why or why not?

Military Service and Earnings (Angrist 1990)

Looking at a cohort of men who were eligible for the Vietnam draft, Angrist uses the draft lottery to construct an instrument for military service. For each cohort, the US government randomly drew birthdays to determine who would be draft-eligible: men whose birthday was drawn early would be drafted before men from the same cohort whose birthday was drawn late. Angrist creates a dummy variable equal to one for those whose birthday was draft-eligible, and zero otherwise, and then uses that as an instrument for military service. He finds, using this IV strategy, that military service lowered eventual earnings by 15%.

Exercise 5. Why do we need an instrument? What might cause military service to be endogenous?

Exercise 6. Do you think that the instrument is as-good-as-randomly assigned? Why or why not?

Military Service and Earnings (Angrist 1990)

Looking at a cohort of men who were eligible for the Vietnam draft, Angrist uses the draft lottery to construct an instrument for military service. For each cohort, the US government randomly drew birthdays to determine who would be draft-eligible: men whose birthday was drawn early would be drafted before men from the same cohort whose birthday was drawn late. Angrist creates a dummy variable equal to one for those whose birthday was draft-eligible, and zero otherwise, and then uses that as an instrument for military service. He finds, using this IV strategy, that military service lowered eventual earnings by 15%.

Exercise 7. What would the reduced form regression be? What is the first stage regression?

Exercise 8. The Vietnam draft had an exemption for college students. Suppose that some young men stayed in college in order to avoid the draft. Would that violate the exclusion restriction? If so, how do you think it might bias the results?

Oregon Health Insurance Experiment (Finkelstein et al., Baicker et al.)

In 2008, Oregon began to expand its Medicaid program (a health insurance program for low-income adults). Since they did not have enough funds to cover everyone who was interested, they decided to randomize: out of 90,000 people on the waitlist, they randomly chose to invite 30,000 people to apply. There was imperfect compliance in both directions: some people in the control group ended up receiving Medicaid, and less than half of the treated group ended up enrolling in Medicaid. However, the treatment increased the probability of enrolling in Medicaid by over 25 percentage points.

A team of researchers used this random assignment to study the effects of Medicaid using an IV. They found that Medicaid improved various financial outcomes, and led to improvements in mental health and wellbeing. However, results for effects on physical health were mixed, and Medicaid receipt increased healthcare spending substantially.

Oregon Health Insurance Experiment (Finkelstein et al., Baicker et al.)

Exercise 9. The mean of a dummy variable for ever being on Medicaid was equal to 0.397 for the treated group, and 0.141 for the control group. What is the first-stage coefficient?

Exercise 10. The mean of a dummy variable for experiencing depression symptoms was equal to 0.306 for the treated group, and 0.329 for the control group. What is the reduced-form coefficient?

Exercise 11. Using the results above, what is the IV estimate of the effect of Medicaid receipt on depression?

Exercise 12. Why do we need an instrument? Would you expect an OLS regression of depression symptoms on Medicaid to recover the causal effect of Medicaid?